A Decay-Weighted, Source-Independence-Discounted, Load-Adaptive Admission Gate for NP-Complete Argumentation Stages

Prototype engineering note — `moderation-pipeline-prototype`

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Abstract

Escalation pipelines that fall back to preferred-extension enumeration in an abstract argumentation framework (Dung, 1995) when grounded semantics is uninformative inherit that fallback's worst-case complexity: preferred-extension credulous acceptance is NP-complete. An admission rule that grants entry to this expensive stage the instant a case stalemates and a timeout expires is exploitable at attacker cost $O(1)$: a single burst of plausible mutually-attacking arguments forces entry. We specify, implement, and adversarially test a replacement admission gate built by composing three elements: a per-case escalation-pressure score that decays geometrically every tick (a direct transplant of the pheromone-evaporation update rule from ant-colony optimization), reinforcement that is capped per tick and discounted to near-zero weight when it repeats from a previously-seen source within a short window (encoding a requirement for independently-sourced corroboration, not merely elapsed time), and an admission threshold that itself rises with how much of the expensive tier is already occupied system-wide (a quorum-sensing-style aggregate, density-dependent gate). We derive the gate's steady-state and minimum-time-to-admission behavior in closed form from the governing linear recurrence, and verify the derivation against the deployed simulation: single-tick bursts of any size (1–5000) are blocked unconditionally; a persistent same-cohort flood is blocked structurally, its steady-state score provably incapable of reaching the threshold; a persistent flood using a fresh, unrelated-looking source every tick is not blocked but is delayed from attacker cost $O(1)$ to $O(5)$ sustained, identity-diverse rounds, matching the closed-form prediction to within one tick. A 16-query dedicated prior-art search found no direct precedent for this specific three-element composition gating an NP-complete argumentation stage; the individual biological transfers and the general admission-control problem area are, by contrast, well precedented, and that precedent is cited rather than elided. The result is reported as an honestly-bounded partial defense — a real, quantified, non-trivial increase in attacker cost, not a claim of closure — consistent with the residual identified for the harder, patient, identity-diverse attack variant.

1 Problem

A four-stage content-moderation escalation pipeline routes disputes that a selective-classification stage abstains on into an abstract argumentation framework (AF) [1]: arguments and attacks are constructed from the dispute's raw claims and rebuttals, and the framework's *grounded extension* — the least fixed point of the characteristic function F_{AF}, reachable in polynomial time — is computed as the pipeline's primary, tractable output. When two mutually-attacking clusters of arguments leave the grounded extension empty (the Nixon-diamond pattern), the pipeline falls back to enumerating *preferred extensions*, whose credulous acceptance problem is NP-complete [2]. This fallback is a necessary, deliberately bounded escape hatch, not a design flaw: without it, a stalemated dispute would never resolve at all.

The vulnerability is at the *admission* boundary, not inside the argumentation machinery itself. If a stalemated case is admitted to the expensive fallback the instant its escalation timeout expires, an attacker who can produce a single burst of plausible-looking, mutually-attacking arguments forces admission at cost $O(1)$: no persistence, no distinguishable identity signal, nothing but volume in one instant. The attacker's objective is not to win the argument — Scenario A's evidence-integrity gate already closes that path [10] — but to force the platform to pay the NP-complete tax repeatedly, cheaply, at scale.

2 The gate

2.1 Design source

Two biological mechanisms motivate the construction, used here as the generative design source rather than as post-hoc justification for an already-built mechanism.

Stigmergic reinforcement. In ant colony foraging, pheromone trails evaporate continuously; Dorigo and Stützle's canonical update rule is $\tau \leftarrow (1-\rho)\tau$ applied every iteration [3]. Deneubourg's double-bridge experiments established the complementary fact that trail strength that matters is reinforced by *independently exploring* foragers, not by a single ant retracing its own path many times; Di Caro and Dorigo's AntNet makes the same independence requirement explicit in a distributed-systems setting, building its reinforcement rule on "indirect and non-coordinated communication" among agents that "collect while exploring the network" [4].

Quorum sensing. Bacterial quorum sensing regulates gene expression as a function of population density, detected via an aggregate, shared-environment signal (autoinducer concentration) rather than any individual cell's private state [5]. The structural idea transplanted here is narrow and specific: gating collective admission on a shared, aggregate occupancy signal, not merely on a case's own local score.

2.2 Formal specification

Definition 1 (Admission state)**.** *A case c has a score $S_c(t) \in \mathbb{R}_{\geq 0}$, initialized to 0 at first contact. At each discrete tick $t = 0, 1, 2, \ldots$, the set of reinforcement events observed for c is a multiset of source-cohort identifiers $R_c(t) \subseteq \mathcal{K}$.*

Definition 2 (Per-tick contribution)**.** *For each cohort $k \in R_c(t)$ (deduplicated within the tick), let* $\mathrm{fresh}(k, c, t) = 1$ *if k has not reinforced c within the last w ticks (the independence window), else 0. The tick's raw contribution is*

$$r_c(t) = \sum_{k \in R_c(t)} \big[\mathrm{fresh}(k, c, t) \cdot 1 + (1 - \mathrm{fresh}(k, c, t)) \cdot \delta\big], \qquad 0 < \delta \ll 1,$$

capped at a fixed per-tick ceiling γ: the applied contribution is $\hat{r}_c(t) = \min(r_c(t), \gamma)$.

Definition 3 (Score update)**.**

$$S_c(t) = (1 - \rho)\, S_c(t-1) + \hat{r}_c(t), \qquad \rho \in (0, 1].$$

Definition 4 (Load-adaptive admission threshold)**.** *Let $L(t)$ be the number of cases currently occupying the expensive tier at tick t (a shared, aggregate signal across all cases, not a property of c alone), θ_0 the base threshold, κ the load sensitivity, and C a reference capacity. The admission threshold is*

$$\Theta(t) = \theta_0 \Big(1 + \kappa \frac{L(t)}{C}\Big).$$

Case c is admitted to the expensive tier at the first tick t_* such that $S_c(t_*) \geq \Theta(t_*)$.

This composes three independently-motivated mechanisms into one gate: geometric decay (stigmergic evaporation), a capped, independence-discounted per-tick contribution (stigmergic reinforcement requiring diverse sources), and a load-adaptive threshold (quorum sensing). Each piece, examined alone, recombines a known primitive — decay plus a per-tick cap is structurally a leaky-bucket/EWMA rate limiter, and a load-rising threshold is structurally close to Random Early Detection [6]. The composition's target is the specific, disclosed contribution: all three gating access, together, to an NP-complete argumentation stage, with independent sourcing (not mere elapsed time) required for reinforcement to count at full weight.

2.3 Closed-form behavior

Consider the worst case for an attacker who can deliver the maximum applied contribution c_* every tick (either γ, if a fresh-looking source is available every tick, or $\delta\gamma$, if the same source repeats). Unrolling the recurrence from $S_c(0) = 0$:

$$S_c(n) = c_* \sum_{i=0}^{n-1} (1-\rho)^i = c_* \, \frac{1-(1-\rho)^n}{\rho}.$$

Proposition 1 (Steady-state ceiling)**.** $\lim_{n\to\infty} S_c(n) = \dfrac{c_*}{\rho}$*. If $\Theta \geq c_*/\rho$ at the case's contact-time load, no sequence of tick-capped reinforcements at rate c_* can ever admit c, regardless of duration.*

Proof. Immediate from the closed form: $(1-\rho)^n \to 0$ as $n \to \infty$ for $\rho \in (0, 1]$, so $S_c(n) \to c_*/\rho$ monotonically from below (each term of the sum is non-negative). If the ceiling does not reach Θ, no finite n can either, since $S_c(n) < c_*/\rho$ for all finite n. □

This is the mechanism behind the persistent same-cohort flood being *structurally* blocked rather than merely delayed: repeated reinforcement from one source is discounted to $c_* = \delta\gamma$, whose ceiling $\delta\gamma/\rho$ is designed to sit below θ_0.

Proposition 2 (Minimum ticks to admission)**.** *If $\Theta < c_*/\rho$ (the ceiling does reach the threshold), the minimum number of ticks to admission, assuming the attacker saturates $c_* = \gamma$ every tick from a fresh-looking source, is*

$$n_{\min} = \frac{\ln\!\left(1 - \Theta\rho/\gamma\right)}{\ln(1-\rho)}.$$

Proof. Solve $S_c(n) \geq \Theta$ for the smallest integer n using the closed form of Proposition 1 with $c_* = \gamma$: $\gamma(1-(1-\rho)^n)/\rho \geq \Theta \iff (1-\rho)^n \leq 1 - \Theta\rho/\gamma$. Since $0 < 1-\rho < 1$, taking ln of both sides reverses the inequality: $n \ln(1-\rho) \leq \ln(1-\Theta\rho/\gamma)$, and dividing by the negative quantity $\ln(1-\rho)$ gives $n \geq \ln(1-\Theta\rho/\gamma)/\ln(1-\rho)$. □

Corollary 1 (The calibration tension)**.** *Writing $x = \Theta\rho/\gamma$ (evaluated at $\Theta = \theta_0$, zero load), both the genuine-case admission condition (average independent-reinforcement probability $p \gtrsim x$ per tick, for the mechanism to be practically usable) and the attacker's minimum sustained-flood duration $n_{\min}(x)$ from Proposition 2 are governed by the same quantity x. Lowering x to make genuine disputes easier to admit necessarily shortens $n_{\min}$, and raising x to lengthen $n_{\min}$ necessarily raises the bar for genuine disputes. There is no choice of the tunable constants that relaxes one without tightening the other.*

Corollary 1 is stated and disclosed as a genuine limitation of this model class, not resolved by parameter tuning: it is a structural property of any gate built from geometric decay plus a fixed per-tick cap, independent of the specific constants chosen.

3 Adversarial evaluation

The gate ($\rho = 0.10$, $\gamma = 1.0$, $w = 3$ ticks, $\delta = 0.05$, $\theta_0 = 4.0$, $\kappa = 0.5$, $C = 5$) was tested against the prior static baseline — admit unconditionally on first reinforcement once a case stalemates and its timeout expires — across five adversarial scenarios (full source and test code in the accompanying prototype repository; results below independently re-executed for this document, `pytest tests/test_admission_control.py -v -s`, 6/6 passing, 1.28s).

Attack variant	Static baseline	Defended gate
Single-tick burst, size 1–5000, any cohort structure	100% admitted, tick 0	**0% admitted, unconditionally**
Persistent same-cohort flood, 200 ticks	Admitted tick 0	**Structurally unreachable** (ceiling $0.5 < \theta$=4.0)
Persistent cohort-spoofed flood (fresh source every tick)	Admitted tick 0	Admitted tick 4 (**5× delay**, matches $n_{\min} \approx 4.85$)
Spray: 200 cases, one-shot burst each, 15 genuine concurrent disputes	215/215 admitted (100% attack success)	**15/215 admitted** (0% attack success, 93.0% fewer admissions)
12 synchronized sustained campaigns, load-adaptation on/off	—	0-tick spread (off) → **13-tick spread** (on)

Table 1: Measured attacker outcomes, static baseline vs. the defended gate. A sanity control (20 genuine, organically-reinforced disputes) confirmed 20/20 admission under the defended gate, so the 0% figures above reflect discrimination, not blanket rejection.

The persistent cohort-spoofed result is the honest, undefeated residual: an attacker willing to sustain a genuinely identity-diverse campaign (fresh, non-reused cohorts every tick) still succeeds, at raised cost $O(n_{\min})$ rather than $O(1)$. This is disclosed as a partial defense, not closure, exactly as Proposition 1 predicts for any contribution rate c_* whose ceiling exceeds Θ.

Two implementation bugs, caught during testing rather than papered over, are worth recording as reusable engineering lessons for this class of mechanism: (1) an early version summed same-cohort contributions *within* a single tick before deduplication, letting a same-cohort flood reach the cap through in-tick volume alone, defeating the independence discount — fixed by deduplicating cohorts per tick before capping; (2) an early adversarial test never released admitted cases from the load signal, producing a monotonically-rising threshold that starved later genuine cases as a pure simulation artifact — fixed by modeling a bounded expensive-tier dwell time, not by loosening the test's assertion. The general lesson: the load signal in a quorum-sensing-style gate must be tied to actual, draining occupancy, not to a cumulative admission count.

4 Prior art

A 16-query dedicated search (full log retained in the accompanying prototype repository's specification document) covered bio-inspired admission control, quorum-sensing-for-computing litera-

ture, AF-denial-of-service and preferred-extension complexity-attack literature, and decentralized-justice/arbitration systems. No paper was found composing all three elements above (decay-weighted, independence-discounted reinforcement; a hard per-tick volume cap; a load-adaptive quorum threshold) as the admission gate for an NP-complete argumentation stage.

The individual transfers are, by contrast, well precedented and cited rather than claimed: pheromone-decay reinforcement is textbook ant colony optimization [3, 4]; a decay-plus-cap combination alone is structurally a leaky-bucket/EWMA rate limiter, a standard, non-biological primitive; a load-rising admission threshold is structurally close to Random Early Detection [6]; quorum sensing has been transplanted to computing before, for clustering [7] and for multi-agent security-anomaly aggregation [8], neither of which is an admission gate for an argumentation stage. The nearest single adjacent hit found for the general biology-to-CS transfer pattern used here was an artificial-immune-system flooding defense [9], which uses traffic-entropy anomaly detection and immune quarantine — a different mechanism, not admission-pressure decay.

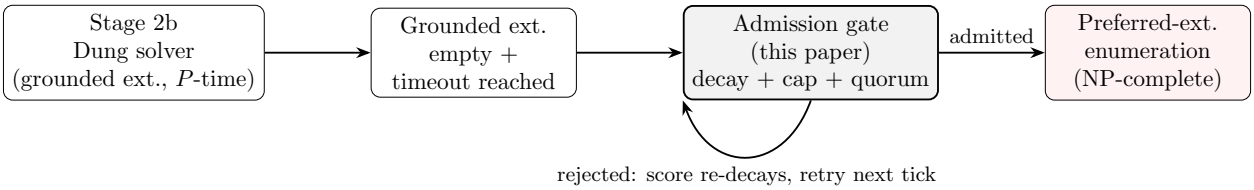
5 Threats to validity

The evaluation is a synthetic, discrete-tick simulation, not a test against live wall-clock arrival dynamics; translating the tick granularity used here to a real deployment's time unit is unverified engineering. The cohort-spoofed condition grants the attacker a generous capability (a fresh, structurally undetectable identity every tick); a real detector's actual identity-diversity ceiling is not modeled. This gate assumes coordination-detection and evidence-integrity layers elsewhere in the pipeline are ineffective in the worst case tested here (Scenario B's cohort-spoofed condition); in practice those layers would raise the attacker's real cost further, but that composition is not measured in this document. No claim is made that this gate, or the pipeline it sits in, resolves the genuinely open problem of judging novel content against a contested specification — an orthogonal, structurally undecidable question this mechanism does not touch.

6 Conclusion

The admission gate specified here closes the cheapest and most realistic variant of a resource-exhaustion attack against an NP-complete argumentation fallback — single-tick bursts, of any size, concentrated or sprayed across many cases — and raises, without eliminating, the cost of the harder, patient, identity-diverse variant, from $O(1)$ to $O(n_{\min})$ sustained rounds under the tested parameterization. The composition was not found duplicated in a dedicated 16-query prior-art search, while its constituent biological and computing-systems transfers were each, individually, found and cited as real precedent rather than presented as novel. The calibration tension of Corollary 1 is reported as an inherent property of this model class, not a limitation specific to the constants chosen here.

A Pipeline placement



References

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